An analytic toy for the shape of $D_\ell^{\rm cross}(x_{\rm HII})$

kSZ^2×galaxy cross-power during reionisation: what sets the curve?

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29 October 2026 (v2: symbols defined, notation cleaned)

1 Question

La Plante, Sipple & Lidz (2022) Fig. 7 plots $D_\ell^{\rm cross}$, the cross-power of a filtered, squared kSZ map with a galaxy map from a redshift window of width $\Delta z = 1$, against the mean ionised fraction $x_{\rm HII}$ of that window. It rises from $x_{\rm HII} \simeq 0$, peaks near $x_{\rm HII} \simeq 0.1$–0.2 and has fallen to almost nothing by $x_{\rm HII} \simeq 0.8$. Is that shape already explained by a few lines of algebra, and which ingredient makes it?

2 Symbols (every symbol, once)

Notation rules used throughout: the *pixel* index is always j; the words "dens" and "ion" (never an index) label the density and ionisation effects; Q is the slice power; χ is a distance; G is the growth factor; D_ℓ is the angular power; R is not used.

Maps and the estimator	
$\Theta(\hat{n})$	the kSZ map: the kSZ temperature fluctuation (units of $T_{\rm CMB}$) summed along the line of sight from all pixels at $z \geq 6$, $\Theta = \sum_j \theta_j$.
θ_j	kSZ contribution of line-of-sight pixel j (1 Mpc long): $\theta_j = -W_j\,(1+\delta_j)\,x_{e,j}\,v_j/c$.
W_j	kSZ weight of pixel j: $W_j = K(z_j)\,\Delta\chi_j$, the optical depth per pixel, damped by $e^{-\tau}$ (see K).
$\delta_j,\ x_{e,j},\ v_j$	matter overdensity, free-electron fraction and line-of-sight peculiar velocity at pixel j; c = speed of light.
$F(\ell)$	the CMB filter, a function of multipole. It keeps the multipoles where the kSZ is not drowned by the primary CMB and noise. $F\Theta$ means: the map Θ with this filter applied in Fourier space.
S	the *squared filtered map*, $S = [F\Theta]^2$. After filtering only small scales remain, so S is the local small-scale kSZ power: bright where the small-scale kSZ is strong.
$g(\hat{n})$	the galaxy (tracer) map of one redshift window: $g = \int dz\, w(z)\, b_g(z)\, \delta_m$, a mean-zero overdensity map.
$w(z)$	window weight, a top hat of width $\Delta z = 1$ with $\int w\, dz = 1$.
$C_\ell^{Sg},\ D_\ell$	C_ℓ^{Sg} is the angular cross-power of S with g; $D_\ell \equiv D_\ell^{\rm cross} = \ell(\ell+1)C_\ell^{Sg}/2\pi$ (times $T_{\rm CMB}^2$ for μK^2; in the toy this is absorbed in A).
Functions of redshift (all ordinary numbers at each z)	
$z,\ \chi(z)$	redshift; comoving distance to z. Multipole and wavenumber are linked by $k = \ell/\chi$.
$\bar{x}(z)$	mean ionised fraction of the universe at z (the reionisation history). $x_{\rm HII}$ of a window is $\bar{x}$ averaged over it.
$\tau(z),\ K(z)$	optical depth to z, $\tau = \int \tau_{\rm pref}\,\bar{x}\,(1+z)^2\,d\chi$; and $K = \tau_{\rm pref}(1+z)^2 e^{-\tau}$, the kSZ weight per unit length ($\tau_{\rm pref} = n_{H,0}\sigma_T$).
$G(z)$	linear growth factor of matter fluctuations, normalised to $G(0) = 1$.
$v(z)$	growth scaling of the peculiar velocity, $v = E\,\Omega_m(z)^{0.55}G/(1+z)$ (= $aHfG$ in units of H_0; $E = H/H_0$). Only its redshift dependence is used.
$b_g(z)$	tracer bias: $\delta_{\rm tracer} \simeq b_g\,\delta_m$ on large scales. $b_g = 2.1(1+z) - 5.3$ (BlueTides fit) for every tracer.
$P_{\rm lin}(k)$	linear matter power spectrum at $z = 0$ (Eisenstein–Hu), so $G^2 P_{\rm lin}$ is the power at z.
$p(z)$	how fast reionisation is going at z: $p = -d\bar{x}/dz \geq 0$.
$b_{zm}(k)$	zreion bias, $b_0/(1+k/k_0)^{0.2}$, $b_0 = 1/1.686$, $k_0 = 0.9\,h{\rm Mpc}^{-1}$. $z_{\rm mean} = 8$ is the reference redshift of the zreion model and $z_{\rm re}(\mathbf{x})$ is the redshift at which the point $\mathbf{x}$ reionises: $z_{\rm re} = z_{\rm mean} + (1+z_{\rm mean})\,b_{zm}\,\delta_L(z_{\rm mean})$.
The toy's own quantities	

$Q(z)$	small-scale kSZ^2 power of the slice of sky at z, *per unit redshift*, at the filter scale (this replaces the symbol s of v1, which clashed with a distance).
δ_L	the long-wavelength (large-scale) matter overdensity of that slice: the part the galaxies trace.
$Q_\delta(z)$	the response of Q to δ_L: $Q_\delta = \partial Q/\partial\delta_L$, i.e. $Q \to Q + Q_\delta\,\delta_L$. It has the same units as Q and is an **ordinary function of z, not an operator**. $Q_\delta = Q_{\rm dens} + Q_{\rm ion}$.
$Q_{\rm dens},\ Q_{\rm ion}$	the two parts of Q_δ: the effect of density (more matter, more electrons) and the effect of ionisation (denser regions reionise earlier). Both are ordinary numbers at each z.
ϵ	ratio of patchy-bubble power to smooth-ionisation power at the filter scale. Fitted ($\simeq 8$).
r_v	small optional correction from the velocity variance; set to 0.
A	overall amplitude per ℓ (fitted; absorbs $T^2_{\rm CMB}$, the filter's share of the power and the velocity normalisation).
$\Delta z,\ z_0$	width ($= 1$) and centre of the galaxy window.

3 The model, in four steps

Step 1: power of one slice. With K, v and $\bar{x}$ as above,

$$Q(z) = K^2 v^2 \left[\bar{x}^2 + \epsilon\,\bar{x}(1-\bar{x})\right]. \tag{1}$$

The first term is the kSZ of a smoothly ionised slice with ionised fraction $\bar{x}$. The second is the extra power of the bubbles: a binary ionisation field has variance $\bar{x}(1-\bar{x})$, which is zero when the slice is fully neutral or fully ionised and largest when half ionised. ϵ sets its size and is the only shape parameter.

Step 2: how Q responds to the large-scale density δ_L.

$$Q_{\rm dens} = (2 + r_v)\,Q, \tag{2}$$

$$Q_{\rm ion} = \frac{\partial Q}{\partial \bar{x}}\,\frac{\partial \bar{x}}{\partial \delta_L}, \qquad \frac{\partial Q}{\partial \bar{x}} = K^2 v^2\left[2\bar{x} + \epsilon(1-2\bar{x})\right], \tag{3}$$

$$\frac{\partial \bar{x}}{\partial \delta_L} = (1 + z_{\rm mean})\,b_{zm}(k)\,\frac{G(z_{\rm mean})}{G(z)}\,p(z). \tag{4}$$

In words: $Q_{\rm dens}$ is the first line because the power goes as $(1+\delta)^2$, which grows by $2\delta_L$ when the slice is denser ($r_v = 0$). $Q_{\rm ion}$ is the change of Q with $\bar{x}$ times the change of $\bar{x}$ with δ_L; the latter is zreion: a denser region reionises earlier ($z_{\rm re}$ shifts up), so at redshift z its ionised fraction is higher by the shift times p. $Q_{\rm dens}$ and $Q_{\rm ion}$ are plain numbers at each z, added together; nothing is applied to anything. The sign of $Q_{\rm ion}$ is that of $2\bar{x} + \epsilon(1-2\bar{x})$, which is positive for $\bar{x} < \epsilon/[2(\epsilon-1)] \simeq 0.57$ and *negative* above it.

Step 3: cross-power with the galaxies. The galaxies trace δ_L with bias b_g. In the Limber approximation only the slice at the same redshift as the galaxies correlates with them (Kramer et al. 2025, Eqs. 9–10, with a tracer in place of τ):

$$D_\ell = \frac{\ell(\ell+1)}{2\pi}\,\frac{A}{\Delta z}\int_{z_0-\Delta z/2}^{z_0+\Delta z/2} dz\,\left(Q_{\rm dens} + Q_{\rm ion}\right) b_g(z)\,\frac{G(z)^2\,P_{\rm lin}(\ell/\chi)}{\chi^2}. \tag{5}$$

The power is taken per unit redshift; the factor $d\chi/dz$ is omitted (including it leaves the best ϵ unchanged, misfit $0.33 \to 0.43$).

Step 4: the reionisation history. La Plante's digitised fiducial $\bar{x}(z)$, made monotonic and Gaussian-smoothed ($\sigma_z = 0.3$) so that p is usable, with $\bar{x} = 1$ below the table and an $e^{-(z-11.86)}$ tail above its end ($\bar{x} = 0.06$).

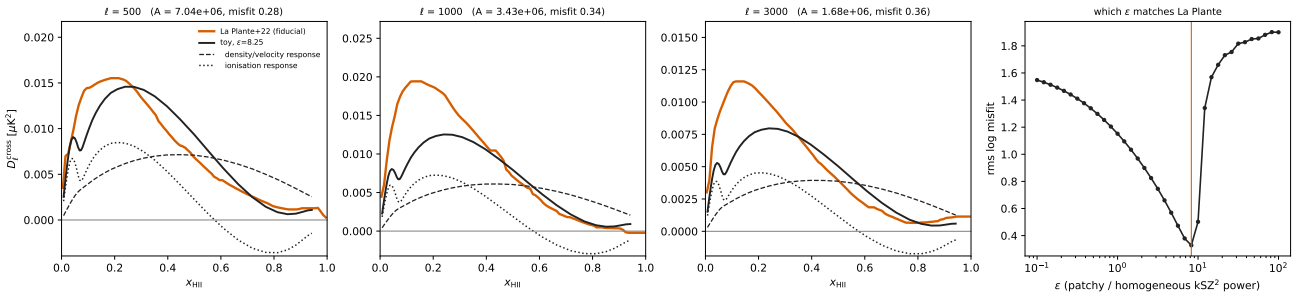
z	6	8	9	10	11	12
$\bar{x}$	0.91	0.44	0.24	0.13	0.08	0.05
τ	0.046	0.062	0.066	0.068	0.070	0.071
K^2v^2 [10^{-10}]	0.42	0.87	1.2	1.6	2.0	2.6
$p = -d\bar{x}/dz$	0.10	0.26	0.15	0.07	0.03	0.04
b_g	9.4	13.6	15.7	17.8	19.9	22.0
$G(z)$	0.18	0.14	0.13	0.12	0.11	0.10

Redshift functions entering the integrand (fiducial history).

4 Fit

For each ϵ and each $\ell \in \{500, 1000, 3000\}$ the amplitude A is the log-space least-squares value against La Plante's digitised Fig. 7 curve over $x_{\rm HII} = 0.1$–0.9 (40 points); the misfit is the rms of $\ln D_{\rm LP} - \ln(A\,D_{\rm toy})$. ϵ is the value

minimising the misfit averaged over the three ℓ.



Orange: La Plante (fiducial). Black: toy ($\epsilon = 8.25$), with its density part ($Q_{\rm dens}$, dashed) and ionisation part ($Q_{\rm ion}$, dotted). Right: misfit versus ϵ. The dotted curve crosses zero near $x_{\rm HII} \simeq 0.57$ and is *negative* above it.

5 Results

- **Best** $\epsilon \simeq 8$ (misfit 0.28/0.34/0.36 in log at $\ell = 500/1000/3000$; $A = 7.0, 3.4, 1.7 \times 10^6$). Peak at $x_{\rm HII} \simeq 0.24$ (La Plante: 0.12–0.19); falls to ~ 0 by $x_{\rm HII} \simeq 0.8$. The shape is therefore reproduced to within the digitisation and the toy's crudeness; the peak sits slightly too late.
- For $\epsilon \lesssim 3$ (smooth term competitive) the misfit is $\gtrsim 0.7$: the peak moves to $x_{\rm HII} \simeq 0.3$ ($\epsilon = 3$), 0.55 ($\epsilon = 1$), 0.7 ($\epsilon = 0$) and the curve keeps 30–70% of its peak value at $x_{\rm HII} = 0.9$, whereas La Plante keeps $\sim 2\%$ (toy at $\epsilon = 8$: 6%).
- The two parts at $\ell = 1000$, relative to the peak of the total ($\epsilon = 8.25$):

$x_{\rm HII}$	0.05	0.13	0.24	0.45	0.72	0.91
$Q_{\rm dens}$ part	0.19	0.32	0.43	0.49	0.37	0.20
$Q_{\rm ion}$ part	0.44	0.49	0.57	0.26	−0.20	−0.14
total	0.63	0.81	1.00	0.75	0.17	0.06

  The density part alone peaks near $x_{\rm HII} \simeq 0.45$ and cannot fall fast enough. The ionisation part peaks near 0.2 and turns *negative* above $\bar{x} \simeq 0.57$: a dense region is then already almost fully ionised, so it has *fewer* bubbles than an average region and less small-scale power. This cancels most of the positive density part, and together with $p \to 0$ as reionisation ends produces the late fall.
- Robustness: $z_{\rm mean} = 7, 8, 9$ give the same ϵ and misfit, because $(1+z_{\rm mean})G(z_{\rm mean})$ is nearly constant. The minimum in ϵ is sharp (misfit 0.33 at 8.25, 0.50 at 10, 1.0 at 11, ≥ 1.3 for $\epsilon \geq 12$) because the total depends on this cancellation; ϵ should not be quoted as a measurement.

6 What it says

The early rise and the late fall of $D_\ell^{\rm cross}(x_{\rm HII})$ are produced by *patchy* ionisation: the kSZ^2 power of a partly ionised slice (growing as $\bar{x}(1-\bar{x})$) is correlated with the large-scale density mainly through where bubbles form (the zreion response, $\propto p(z)$), not through the smooth density term. This is consistent with the component study, where $v_{\rm los}$ and the ionisation morphology both follow the kSZ curve. A tracer that is a better proxy for δ_L at the patchy epoch (larger effective b_g, less noise) therefore carries more of this signal.

7 Caveats and checks to do

- Squeezed-limit, slice-diagonal toy: no cross-slice pairs, no line-of-sight cancellation of the smooth term. $\epsilon \sim 8$ may partly stand in for that cancellation, so it is a statement about the relative size of the patchy and smooth terms in the *estimator*, not a measured physical ratio.
- $x_{\rm HII} < 0.1$ is controlled by the first points of the digitised history (a small wiggle remains); $b_g(z)$ is one shape for all tracers; only the shape is fitted, the amplitude is not predicted.
- Not yet done: the same overlay on our own measured curves (`--ours-root`) and a direct test of ϵ from the simulation (power of the x_e-only versus v-only kSZ^2 maps per window, scripts 48/49).

8 Reproduce

```
python scripts/50_toy_model.py --out-dir data/plots/2026-10-23_toy \
   [--ours-root data/plots/2026-10-16_tracer_cross] [--eps 8.25]
```